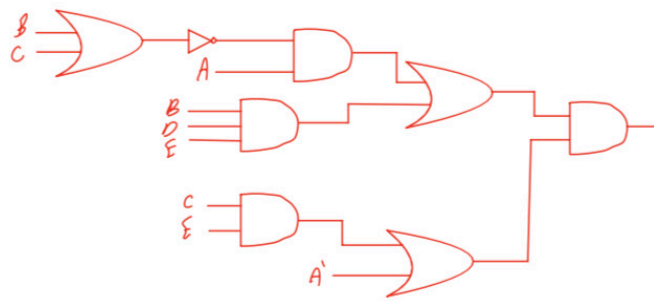


1- Draw the circuit of the function

$$F = [A(B + C)' + BDE] (A' + CE) :$$



2- Design a circuit that has a 3- bit binary input and produce a single output. The output function $f(x,y,z) = 1$ whenever $x+y = y+z$ 0 otherwise.



d- Show the truth table for the previous combinational circuit :

x	y	z	F
0	0	0	1
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	1

e- Determine the minimize POS for the output function:

$$f' = x'y'z + x'yz + xy'z' + xyz'$$

$$f' = x'z(y' + y) + xz'(y' + y) = x'z + xz' = x \oplus z$$

$$f'' = f = (x'z)' \cdot (xz')'$$
$$= (x + z') \cdot (x' + z)$$

f- Draw the minimize circuit diagram using only one gate. Assume both the normal and complement inputs are available.

